

Artificial intelligence Research Report

Generated on 11 Jun 2026, 04:48 PM

Executive Summary

Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. It is a field of research in engineering, mathematics and computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals. High-profile applications of AI include advanced web search engines, chatbots, virtual assistants, autonomous vehicles, and play and analysis in strategy games (e.g., chess and Go). Since the 2020s, generative AI has become widely available to generate images, audio, and videos from text prompts. The traditional goals of AI research include learning, reasoning, knowledge representation, planning, natural language processing, and perception, as well as support for robotics. To reach these goals, AI researchers have used techniques including state space search and mathematical optimization, formal logic, artificial neural networks, and methods based on statistics, operations research, and economics. AI also draws upon psychology, linguistics, philosophy, neuroscience, and other fields. Some companies, such as OpenAI, Google DeepMind and Meta, aim to create artificial general intelligence (AGI) – AI that can complete virtually any cognitive task at least as well as a human. Artificial intelligence was founded as an academic discipline in 1956, and the field went through multiple cycles of optimism throughout its history, followed by periods of disappointment and loss of funding, known as AI winters. Funding and interest increased substantially after 2012, when graphics processing units began being used to accelerate neural networks, and deep learning outperformed previous AI techniques. This growth accelerated further after 2017 with the transformer architecture. In the 2020s, an AI boom has coincided with advances in generative AI, which allowed for the creation and modification of media. In addition to AI safety and unintended consequences and harms from the use of AI, ethical concerns, AI's long-term effects, and potential existential risks have prompted discussions of AI regulation.

- Reference topic identified as Artificial intelligence.
- 8 background sections were collected.
- 2 current news item(s) were collected.

Source Status

All configured sources responded successfully.

Background And Context

Source: https://en.wikipedia.org/wiki/Artificial_intelligence

Goals

The general problem of simulating (or creating) intelligence has been broken into subproblems. These consist of particular traits or capabilities that researchers expect an intelligent system to display. The traits described below have received the most attention and cover the scope of AI research.

- Reasoning and problem-solving: Early researchers developed algorithms that imitated step-by-step reasoning that humans use when they solve puzzles or make logical deductions. By the late 1980s and 1990s, methods were developed for dealing with uncertain or incomplete information, employing concepts from probability and economics. Many of these algorithms are insufficient for solving large reasoning problems because they experience a "combinatorial explosion": They become exponentially slower as the problems grow. Even humans rarely use the step-by-step deduction that early AI research could model. Humans solve most of their problems using fast, intuitive judgments. Reasoning models, a type of large language model trained to generate intermediate chains-of-thought, emerged in 2024 and allowed improved performance on complex problems in mathematics and coding.
- Knowledge representation: Knowledge representation and knowledge engineering allow AI programs to answer questions intelligently and make deductions about real-world facts. Formal knowledge representations are used in content-based indexing and retrieval, scene interpretation, clinical decision support, knowledge discovery (mining "interesting" and actionable inferences from large databases), and other areas. A knowledge base is a body of knowledge represented in a form that can be used by a program. An ontology is the set of objects, relations, concepts, and properties used by a particular domain of knowledge. Knowledge bases need to represent things such as objects, properties, categories, and relations between objects; situations, events, states, and time; causes and effects; knowledge about knowledge (what we know about what other people know); default reasoning (things that humans assume are true until they are told differently and will remain true even when other facts are changing); and many other aspects and domains of knowledge. Among the most difficult problems in knowledge representation are the breadth of commonsense knowledge (the set of atomic facts that the average person knows is enormous); and the sub-symbolic form of most commonsense knowledge (much of what people know is not represented as "facts" or "statements" that they could express verbally). There is also the difficulty of knowledge acquisition, the problem of obtaining knowledge for AI applications.
- Planning and decision-making: An "agent" is any entity (artificial or not) that perceives and takes actions in the world. A rational agent has goals or preferences and takes actions to make them happen. In automated planning, the agent has a specific goal. In automated decision-making, the agent has preferences—there are some situations it would prefer to be in, and some situations it is trying to avoid. The decision-making agent assigns a number to each situation (called its "utility") that measures how much the agent prefers it. For each possible action, it can calculate the "expected utility": the utility of all possible outcomes of the action, weighted by the probability that the outcome will occur. It can then choose the action with the maximum expected utility. In classical planning, the agent knows exactly what the effect of any action will be. In most real-world problems, however, the agent may not be certain about the situation it is in (it is "unknown" or "unobservable") and it may not know for certain what will happen after each possible action (it is not "deterministic"). It must choose an action by making a probabilistic guess and then reassess the situation to see if the action worked. Alongside thorough testing and improvement based on previous decisions, having an explanation for why the agent took certain decisions is a way to build trust, especially when the decisions have to be relied upon. In some problems, the agent's preferences may be uncertain, especially if there are other agents or humans involved. These preferences be learned (e.g., with inverse reinforcement learning), or the agent can seek information to improve them. Information value theory can be used to weigh the value of exploratory or experimental actions. The space of possible future actions and situations is typically intractably large, so the agents must take actions and evaluate situations while being uncertain of what the outcome will be. A Markov decision process has a transition model that describes the probability that a particular action will change the state in a particular way and a reward function that supplies the utility of each state and the cost of each action. A policy associates a decision with each possible state. The policy could be calculated (e.g., by iteration), be heuristic, or it can be learned. Game

theory describes the rational behavior of multiple interacting agents and is used in AI programs that make decisions that involve other agents.

- **Learning:** Machine learning is the study of programs that can improve their performance on a given task automatically. It has been a part of AI from the beginning. There are several kinds of machine learning. Unsupervised learning analyzes a stream of data and finds patterns and makes predictions without any other guidance. Supervised learning requires labeling the training data with the expected answers, and comes in two main varieties: classification (where the program must learn to predict what category the input belongs in) and regression (where the program must deduce a numeric function based on numeric input). In reinforcement learning, the agent is rewarded for good responses and punished for bad ones. The agent learns to choose responses that are classified as "good". Transfer learning is when the knowledge gained from one problem is applied to a new problem. Deep learning is a type of machine learning that runs inputs through biologically inspired artificial neural networks for all of these types of learning. Computational learning theory can assess learners by computational complexity, by sample complexity (how much data is required), or by other notions of optimization.
- **Natural language processing:** Natural language processing (NLP) allows programs to read, write and communicate in human languages. Specific problems include speech recognition, speech synthesis, machine translation, information extraction, information retrieval and question answering. Early work, based on Noam Chomsky's generative grammar and semantic networks, had difficulty with word-sense disambiguation unless restricted to small domains called "micro-worlds" (due to the common sense knowledge problem). Margaret Masterman believed that it was meaning and not grammar that was the key to understanding languages, and that thesauri and not dictionaries should be the basis of computational language structure. Modern deep learning techniques for NLP include word embedding (representing words, typically as vectors encoding their meaning), transformers (a deep learning architecture using an attention mechanism), and others. In 2019, generative pre-trained transformer (or "GPT") language models began to generate coherent text, and by 2023, these models were able to get human-level scores on the bar exam, SAT test, GRE test, and many other real-world applications.

Techniques

AI research uses a wide variety of techniques to accomplish the goals above.

- Search and optimization: There are two different kinds of search used in AI: state space search and local search:
- Logic: Formal logic is used for reasoning and knowledge representation. Formal logic comes in two main forms: propositional logic (which operates on statements that are true or false and uses logical connectives such as "and", "or", "not" and "implies") and predicate logic (which also operates on objects, predicates and relations and uses quantifiers such as "Every X is a Y" and "There are some Xs that are Ys"). Deductive reasoning in logic is the process of proving a new statement (conclusion) from other statements that are given and assumed to be true (the premises). Proofs can be structured as proof trees, in which nodes are labelled by sentences, and children nodes are connected to parent nodes by inference rules. Given a problem and a set of premises, problem-solving reduces to searching for a proof tree whose root node is labelled by a solution of the problem and whose leaf nodes are labelled by premises or axioms. In the case of Horn clauses, problem-solving search can be performed by reasoning forwards from the premises or backwards from the problem. In the more general case of the clausal form of first-order logic, resolution is a single, axiom-free rule of inference, in which a problem is solved by proving a contradiction from premises that include the negation of the problem to be solved. Inference in both Horn clause logic and first-order logic is undecidable, and therefore intractable. However, backward reasoning with Horn clauses, which underpins computation in the logic programming language Prolog, is Turing complete. Moreover, its efficiency is competitive with computation in other symbolic programming languages. Fuzzy logic assigns a "degree of truth" between 0 and 1. It can therefore handle propositions that are vague and partially true. Non-monotonic logics, including logic programming with negation as failure, are designed to handle default reasoning. Other specialized versions of logic have been developed to describe many complex domains.
- Probabilistic methods for uncertain reasoning: Many problems in AI (including reasoning, planning, learning, perception, and robotics) require the agent to operate with incomplete or uncertain information. AI researchers have devised a number of tools to solve these problems using methods from probability theory and economics. Precise mathematical tools have been developed that analyze how an agent can make choices and plan, using decision theory, decision analysis, and information value theory. These tools include models such as Markov decision processes, dynamic decision networks, game theory and mechanism design. Bayesian networks are a tool that can be used for reasoning (using the Bayesian inference algorithm), learning (using the expectation–maximization algorithm), planning (using decision networks) and perception (using dynamic Bayesian networks). Probabilistic algorithms can also be used for filtering, prediction, smoothing, and finding explanations for streams of data, thus helping perception systems analyze processes that occur over time (e.g., hidden Markov models or Kalman filters).
- Classifiers and statistical learning methods: The simplest AI applications can be divided into two types: classifiers (e.g., "if shiny then diamond"), on one hand, and controllers (e.g., "if diamond then pick up"), on the other hand. Classifiers are functions that use pattern matching to determine the closest match. They can be fine-tuned based on chosen examples using supervised learning. Each pattern (also called an "observation") is labeled with a certain predefined class. All the observations combined with their class labels are known as a data set. When a new observation is received, that observation is classified based on previous experience. There are many kinds of classifiers in use. The decision tree is the simplest and most widely used symbolic machine learning algorithm. K-nearest neighbor algorithm was the most widely used analogical AI until the mid-1990s, and Kernel methods such as the support vector machine (SVM) displaced k-nearest neighbor in the 1990s. The naive Bayes classifier is reportedly the "most widely used learner" at Google, due in part to its scalability. Neural networks are also used as classifiers.
- Artificial neural networks: An artificial neural network is based on a collection of nodes also known as artificial neurons, which loosely model the neurons in a biological brain. It is trained to recognise patterns; once trained, it can recognise those patterns in fresh data. There is an input, at least one hidden layer of nodes and an output. Each node applies a function and once the weight crosses its specified threshold, the

data is transmitted to the next layer. A network is typically called a deep neural network if it has at least 2 hidden layers. Learning algorithms for neural networks use local search to choose the weights that will get the right output for each input during training. The most common training technique is the backpropagation algorithm. Neural networks learn to model complex relationships between inputs and outputs and find patterns in data. In theory, a neural network can learn any function. In feedforward neural networks the signal passes in only one direction. The term perceptron typically refers to a single-layer neural network. In contrast, deep learning uses many layers. Recurrent neural networks (RNNs) feed the output signal back into the input, which allows short-term memories of previous input events. Long short-term memory networks (LSTMs) are recurrent neural networks that better preserve longterm dependencies and are less sensitive to the vanishing gradient problem. Convolutional neural networks (CNNs) use layers of kernels to more efficiently process local patterns. This local processing is especially important in image processing, where the early CNN layers typically identify simple local patterns such as edges and curves, with subsequent layers detecting more complex patterns like textures, and eventually whole objects.

Applications

AI and machine learning technology is used in most of the essential applications of the 2020s, including: search engines (such as Google Search) targeting online advertisements recommendation systems (offered by Netflix, YouTube or Amazon) driving internet traffic targeted advertising (AdSense, Facebook) virtual assistants (such as Siri or Alexa) autonomous vehicles (including drones, ADAS and self-driving cars) automatic language translation (Microsoft Translator, Google Translate) facial recognition (Apple's FaceID or Microsoft's DeepFace and Google's FaceNet) image labeling (used by Facebook, Apple's Photos and TikTok). The deployment of AI may be overseen by a chief automation officer (CAO).

- **Health and medicine:** It has been suggested that AI can overcome discrepancies in funding allocated to different fields of research. AlphaFold 2 (2021) demonstrated the ability to approximate, in hours rather than months, the 3D structure of a protein. In 2023, it was reported that AI-guided drug discovery helped find a class of antibiotics capable of killing two different types of drug-resistant bacteria. In 2024, researchers used machine learning to accelerate the search for Parkinson's disease drug treatments. Their aim was to identify compounds that block the clumping, or aggregation, of alpha-synuclein (the protein that characterises Parkinson's disease). They were able to speed up the initial screening process ten-fold and reduce the cost by a thousand-fold.
- **Gaming:** Game playing programs have been used since the 1950s to demonstrate and test AI's most advanced techniques. Deep Blue became the first computer chess-playing system to beat a reigning world chess champion, Garry Kasparov, on 11 May 1997. In 2011, in a Jeopardy! quiz show exhibition match, IBM's question answering system, Watson, defeated the two greatest Jeopardy! champions, Brad Rutter and Ken Jennings, by a significant margin. In March 2016, AlphaGo won 4 out of 5 games of Go in a match with Go champion Lee Sedol, becoming the first computer Go-playing system to beat a professional Go player without handicaps. Then, in 2017, it defeated Ke Jie, who was the best Go player in the world. Other programs handle imperfect-information games, such as the poker-playing program Pluribus. DeepMind developed increasingly generalistic reinforcement learning models, such as with MuZero, which could be trained to play chess, Go, or Atari games. In 2019, DeepMind's AlphaStar achieved grandmaster level in StarCraft II, a particularly challenging real-time strategy game that involves incomplete knowledge of what happens on the map. In 2021, an AI agent competed in a PlayStation Gran Turismo competition, winning against four of the world's best Gran Turismo drivers using deep reinforcement learning. In 2024, Google DeepMind introduced SIMA, a type of AI capable of autonomously playing nine previously unseen open-world video games by observing screen output, as well as executing short, specific tasks in response to natural language instructions.
- **Mathematics:** In mathematics, probabilistic large language models are versatile, but can also produce wrong answers in the form of hallucinations. The Alibaba Group developed a version of its Qwen models called Qwen2-Math, that achieved state-of-the-art performance on several mathematical benchmarks, including 84% accuracy on the MATH dataset of competition mathematics problems. In January 2025, Microsoft proposed the technique rStar-Math that leverages Monte Carlo tree search and step-by-step reasoning, enabling a relatively small language model like Qwen-7B to solve 53% of the AIME 2024 and 90% of the MATH benchmark problems. Google DeepMind has developed models for solving mathematical problems: AlphaTensor, AlphaGeometry, AlphaProof, AlphaEvolve, and FunSearch. When natural language is used to describe mathematical problems, converters can transform such prompts into a formal language such as Lean to define mathematical tasks. The experimental model Gemini Deep Think accepts natural language prompts directly and achieved gold medal results in the International Math Olympiad of 2025. Topological deep learning integrates various topological approaches.
- **Finance:** According to Nicolas Firzli, director of the World Pensions & Investments Forum, it may be too early to see the emergence of highly innovative AI-informed financial products and services. He argues that "the deployment of AI tools will simply further automatise things: destroying tens of thousands of jobs in banking, financial planning, and pension advice in the process, but I'm not sure it will unleash a new wave of [e.g., sophisticated] pension innovation."

- Military: Various countries are deploying AI military applications. The main applications enhance command and control, communications, sensors, integration and interoperability. Research is targeting intelligence collection and analysis, logistics, cyber operations, information operations, and semiautonomous and autonomous vehicles. AI technologies enable coordination of sensors and effectors, threat detection and identification, marking of enemy positions, target acquisition, coordination and deconfliction of distributed Joint Fires between networked combat vehicles, both human-operated and autonomous. AI has been used in military operations in Iraq, Syria, Israel and Ukraine.

Ethics

AI has potential benefits and potential risks. AI may be able to advance science and find solutions for serious problems: Demis Hassabis of DeepMind hopes to "solve intelligence, and then use that to solve everything else". However, as the use of AI has become widespread, several unintended consequences and risks have been identified. In-production systems can sometimes not factor ethics and bias into their AI training processes, especially when the AI algorithms are inherently unexplainable in deep learning.

- **Ethical machines and alignment:** Friendly AI are machines that have been designed from the beginning to minimize risks and to make choices that benefit humans. Eliezer Yudkowsky, who coined the term, argues that developing friendly AI should be a higher research priority: it may require a large investment and it must be completed before AI becomes an existential risk. Machines with intelligence have the potential to use their intelligence to make ethical decisions. The field of machine ethics provides machines with ethical principles and procedures for resolving ethical dilemmas. The field of machine ethics is also called computational morality, and was founded at an AAAI symposium in 2005. Other approaches include Wendell Wallach's "artificial moral agents" and Stuart J. Russell's three principles for developing provably beneficial machines.
- **Open source:** Active organizations in the AI open-source community include Hugging Face, Google, EleutherAI and Meta. Various AI models, such as Llama 2, Mistral or Stable Diffusion, have been made open-weight, meaning that their architecture and trained parameters (the "weights") are publicly available. Open-weight models can be freely fine-tuned, which allows companies to specialize them with their own data and for their own use-case. Open-weight models are useful for research and innovation but can also be misused. Since they can be fine-tuned, any built-in security measure, such as objecting to harmful requests, can be trained away until it becomes ineffective. Some researchers warn that future AI models may develop dangerous capabilities (such as the potential to drastically facilitate bioterrorism) and that once released on the Internet, they cannot be deleted everywhere if needed. They recommend pre-release audits and cost-benefit analyses.
- **Frameworks:** Artificial intelligence projects can be guided by ethical considerations during the design, development, and implementation of an AI system. An AI framework such as the Care and Act Framework, developed by the Alan Turing Institute and based on the SUM values, outlines four main ethical dimensions, defined as follows: Respect the dignity of individual people Connect with other people sincerely, openly, and inclusively Care for the wellbeing of everyone Protect social values, justice, and the public interest Other developments in ethical frameworks include those decided upon during the Asilomar Conference, the Montreal Declaration for Responsible AI, and the IEEE's Ethics of Autonomous Systems initiative, among others; however, these principles are not without criticism, especially regarding the people chosen to contribute to these frameworks. Promotion of the wellbeing of the people and communities that these technologies affect requires consideration of the social and ethical implications at all stages of AI system design, development and implementation, and collaboration between job roles such as data scientists, product managers, data engineers, domain experts, and delivery managers. The UK AI Safety Institute released in 2024 a testing toolset called 'Inspect' for AI safety evaluations available under an MIT open-source licence which is freely available on GitHub and can be improved with third-party packages. It can be used to evaluate AI models in a range of areas including core knowledge, ability to reason, and autonomous capabilities.
- **Regulation:** The regulation of artificial intelligence is the development of public sector policies and laws for promoting and regulating AI; it is therefore related to the broader regulation of algorithms. The regulatory and policy landscape for AI is an emerging issue in jurisdictions globally. According to AI Index at Stanford, the annual number of AI-related laws passed in the 127 survey countries jumped from one passed in 2016 to 37 passed in 2022 alone. Between 2016 and 2020, more than 30 countries adopted dedicated strategies for AI. Most EU member states had released national AI strategies, as had Canada, China, India, Japan, Mauritius, the Russian Federation, Saudi Arabia, United Arab Emirates, U.S., and Vietnam. Others were in the process of elaborating their own AI strategy, including Bangladesh, Malaysia and Tunisia. The Global Partnership on Artificial Intelligence was launched in June 2020, stating a need for AI to be developed in

accordance with human rights and democratic values, to ensure public confidence and trust in the technology. Henry Kissinger, Eric Schmidt, and Daniel Huttenlocher published a joint statement in November 2021 calling for a government commission to regulate AI. In 2023, OpenAI leaders published recommendations for the governance of superintelligence, which they believe may happen in less than 10 years. In 2023, the United Nations also launched an advisory body to provide recommendations on AI governance; the body comprises technology company executives, government officials and academics. On 1 August 2024, the EU Artificial Intelligence Act entered into force, establishing the first comprehensive EU-wide AI regulation. In 2024, the Council of Europe created the first international legally binding treaty on AI, called the "Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law". It was adopted by the European Union, the United States, the United Kingdom, and other signatories. In a 2022 Ipsos survey, attitudes towards AI varied greatly by country; 78% of Chinese citizens, but only 35% of Americans, agreed that "products and services using AI have more benefits than drawbacks". A 2023 Reuters/Ipsos poll found that 61% of Americans agree, and 22% disagree, that AI poses risks to humanity. In a 2023 Fox News poll, 35% of Americans thought it "very important", and an additional 41% thought it "somewhat important", for the federal government to regulate AI, versus 13% responding "not very important" and 8% responding "not at all important". In November 2023, the first global AI Safety Summit was held in Bletchley Park in the UK to discuss the near and far term risks of AI and the possibility of mandatory and voluntary regulatory frameworks. 28 countries including the United States, China, and the European Union issued a declaration at the start of the summit, calling for international co-operation to manage the challenges and risks of artificial intelligence. In May 2024 at the AI Seoul Summit, 16 global AI tech companies agreed to safety commitments on the development of AI. In March 2026, the United Nations convened the inaugural meeting of the Independent International Scientific Panel on AI, a 40-member expert body established under the Global Digital Compact to produce annual evidence-based reports on AI's societal impacts.

History

The study of mechanical or "formal" reasoning began with philosophers and mathematicians in antiquity. The study of logic led directly to Alan Turing's theory of computation, which suggested that a machine, by shuffling symbols as simple as "0" and "1", could simulate any conceivable form of mathematical reasoning. This, along with concurrent discoveries in cybernetics, information theory and neurobiology, led researchers to consider the possibility of building an "electronic brain". They developed several areas of research that would become part of AI, such as McCulloch and Pitts design for "artificial neurons" in 1943, and Turing's influential 1950 paper 'Computing Machinery and Intelligence', which introduced the Turing test and showed that "machine intelligence" was plausible. The field of AI research was founded at a workshop at Dartmouth College in 1956. The first AI program, Logic Theorist, was presented at the workshop, created by future Turing Award winner Allen Newell and future Nobel Laureate Herbert A. Simon, in collaboration with J. C. Shaw. Many of the workshop attendees became the leaders of AI research in the 1960s. They and their students produced programs that the press described as "astonishing": computers were learning checkers strategies, solving word problems in algebra, proving logical theorems and speaking English. Artificial intelligence laboratories were set up at a number of British and U.S. universities in the latter 1950s and early 1960s. Researchers in the 1960s and the 1970s were convinced that their methods would eventually succeed in creating a machine with general intelligence and considered this the goal of their field. In 1965 Herbert Simon predicted, "machines will be capable, within twenty years, of doing any work a man can do". In 1967 Marvin Minsky agreed, writing that "within a generation ... the problem of creating 'artificial intelligence' will substantially be solved". They had, however, underestimated the difficulty of the problem. In 1974, both the U.S. and British governments cut off exploratory research in response to the criticism of Sir James Lighthill and ongoing pressure from the U.S. Congress to fund more productive projects. Minsky and Papert's book *Perceptrons* was understood as proving that artificial neural networks would never be useful for solving real-world tasks, thus discrediting the approach altogether. The "AI winter", a period when obtaining funding for AI projects was difficult, followed. In the early 1980s, AI research was revived by the commercial success of expert systems, a form of AI program that simulated the knowledge and analytical skills of human experts. By 1985, the market for AI had reached over a billion dollars. At the same time, Japan's fifth generation computer project inspired the U.S. and British governments to restore funding for academic research. However, beginning with the collapse of the Lisp Machine market in 1987, AI once again fell into disrepute, and a second, longer-lasting winter began. Up to this point, most of AI's funding had gone to projects that used high-level symbols to represent mental objects like plans, goals, beliefs, and known facts. In the 1980s, some researchers began to doubt that this approach would be able to imitate all the processes of human cognition, especially perception, robotics, learning and pattern recognition, and began to look into "sub-symbolic" approaches. Rodney Brooks rejected "representation" in general and focussed directly on engineering machines that move and survive. Judea Pearl, Lotfi Zadeh, and others developed methods that handled incomplete and uncertain information by making reasonable guesses rather than precise logic. But the most important development was the revival of "connectionism", including neural network research, by Geoffrey Hinton and others. In 1990, Yann LeCun successfully showed that convolutional neural networks can recognize handwritten digits, the first of many successful applications of neural networks. AI gradually restored its reputation in the late 1990s and early 21st century by exploiting formal mathematical methods and by finding specific solutions to specific problems. This "narrow" and "formal" focus allowed researchers to produce verifiable results and collaborate with other fields (such as statistics, economics and mathematics). By 2000, solutions developed by AI researchers were being widely used, although in the 1990s they were rarely described as "artificial intelligence" (a tendency known as the AI effect). However, several academic researchers became concerned that AI was no longer pursuing its original goal of creating versatile, fully intelligent machines. Beginning around 2002, they founded the subfield of artificial general intelligence (or "AGI"), which had several well-funded institutions by the 2010s. Deep learning began to dominate industry benchmarks in 2012 and was adopted throughout the field. For many specific tasks, other methods were abandoned. Deep learning's success was based on both hardware improvements (faster computers, graphics processing units, cloud computing) and access to large amounts of data (including curated datasets, such as ImageNet). Deep learning's success led to an enormous increase in interest and funding in AI. The amount of machine learning research (measured by total publications)

increased by 50% in the years 2015–2019. In 2016, issues of fairness and the misuse of technology were catapulted into center stage at machine learning conferences, publications vastly increased, funding became available, and many researchers re-focussed their careers on these issues. The alignment problem became a serious field of academic study. In the late 2010s and early 2020s, AGI companies began to deliver programs that created enormous interest. In 2015, AlphaGo, developed by DeepMind, beat the world champion Go player. The program taught only the game's rules and developed a strategy by itself. GPT-3 is a large language model that was released in 2020 by OpenAI and is capable of generating high-quality human-like text. ChatGPT, launched on 30 November 2022, became the fastest-growing consumer software application in history, gaining over 100 million users in two months. It marked what is widely regarded as AI's breakout year, bringing it into the public consciousness. These programs, and others, inspired an aggressive AI boom, where large companies began investing billions of dollars in AI research. According to AI Impacts, about US\$50 billion annually was invested in "AI" around 2022 in the U.S. alone and about 20% of the new U.S. computer science PhD graduates have specialized in "AI". About 800,000 "AI"-related U.S. job openings existed in 2022. According to PitchBook research, 22% of newly funded startups in 2024 claimed to be AI companies.

Philosophy

Philosophical debates have historically sought to determine the nature of intelligence and how to make intelligent machines. Another major focus has been whether machines can be conscious, and the associated ethical implications. Many other topics in philosophy are relevant to AI, such as epistemology and free will. Rapid advancements have intensified public discussions on the philosophy and ethics of AI.

- **Defining artificial intelligence:** Alan Turing investigated whether machines can show intelligent behaviour and think. In 1950, he proposed the Turing test, which measures the ability of a machine to simulate human conversation. Since we can only observe the behavior of the machine, it does not matter if it is "actually" thinking or literally has a "mind". Turing notes that we can not determine these things about other people but "it is usual to have a polite convention that everyone thinks." Russell and Norvig agree with Turing that intelligence must be defined in terms of external behavior, not internal structure. However, they are critical that the test requires the machine to imitate humans. "Aeronautical engineering texts", they wrote, "do not define the goal of their field as making 'machines that fly so exactly like pigeons that they can fool other pigeons.'" AI founder John McCarthy agreed, writing that "Artificial intelligence is not, by definition, simulation of human intelligence". McCarthy defines intelligence as "the computational part of the ability to achieve goals in the world". Another AI founder, Marvin Minsky, similarly describes it as "the ability to solve hard problems". Artificial Intelligence: A Modern Approach defines it as the study of agents that perceive their environment and take actions that maximize their chances of achieving defined goals. The many differing definitions of AI have been critically analyzed. During the 2020s AI boom, the term has been used as a marketing buzzword to promote products and services which do not use AI.
- **Evaluating approaches to AI:** No established unifying theory or paradigm has guided AI research for most of its history. The unprecedented success of statistical machine learning in the 2010s eclipsed all other approaches (so much so that some sources, especially in the business world, use the term "artificial intelligence" to mean "machine learning with neural networks"). This approach is mostly sub-symbolic, soft and narrow.
- **Machine consciousness, sentience, and mind:** There is no settled consensus in philosophy of mind on whether a machine can have a mind, consciousness and mental states in the same sense that human beings do. This issue considers the internal experiences of the machine, rather than its external behavior. Mainstream AI research considers this issue irrelevant because it does not affect the goals of the field: to build machines that can solve problems using intelligence. Russell and Norvig add that "[t]he additional project of making a machine conscious in exactly the way humans are is not one that we are equipped to take on." However, the question has become central to the philosophy of mind. It is also typically the central question at issue in artificial intelligence in fiction.

Future

- **Superintelligence and the singularity:** A superintelligence is a hypothetical agent that would possess intelligence far surpassing that of the brightest and most gifted human mind. If research into artificial general intelligence produced sufficiently intelligent software, it might be able to reprogram and improve itself. The improved software would be even better at improving itself, leading to what I. J. Good called an "intelligence explosion" and Vernor Vinge called a "singularity". However, technologies cannot improve exponentially indefinitely, and typically follow an S-shaped curve, slowing when they reach the physical limits of what the technology can do.
- **Transhumanism:** Robot designer Hans Moravec, cyberneticist Kevin Warwick and inventor Ray Kurzweil have predicted that humans and machines may merge in the future into cyborgs that are more capable and powerful than either. This idea, called transhumanism, has roots in the writings of Aldous Huxley and Robert Ettinger. Edward Fredkin argues that "artificial intelligence is the next step in evolution", an idea first proposed by Samuel Butler's "Darwin among the Machines" as far back as 1863, and expanded upon by George Dyson in his 1998 book *Darwin Among the Machines: The Evolution of Global Intelligence*.

In fiction

Thought-capable artificial beings have appeared as storytelling devices since antiquity, and have been a persistent theme in science fiction. A common trope in these works began with Mary Shelley's *Frankenstein*, where a human creation becomes a threat to its masters. This includes such works as Arthur C. Clarke's and Stanley Kubrick's *2001: A Space Odyssey* (both 1968), with HAL 9000, the murderous computer in charge of the *Discovery One* spaceship, as well as *Blade Runner* (1982), *The Terminator* (1984) and *The Matrix* (1999). In contrast, the rare loyal robots such as Gort from *The Day the Earth Stood Still* (1951) and Bishop from *Aliens* (1986) are less prominent in popular culture. Isaac Asimov introduced the Three Laws of Robotics in many stories, most notably with the "Multivac" super-intelligent computer. Asimov's laws are often brought up during lay discussions of machine ethics; while almost all artificial intelligence researchers are familiar with Asimov's laws through popular culture, they generally consider the laws useless for many reasons, one of which is their ambiguity. Several works use AI to force us to confront the fundamental question of what makes us human, showing us artificial beings that have the ability to feel, and thus to suffer. This appears in Karel Čapek's *R.U.R.*, the films *A.I. Artificial Intelligence* and *Ex Machina*, as well as the novel *Do Androids Dream of Electric Sheep?*, by Philip K. Dick. Dick considers the idea that our understanding of human subjectivity is altered by technology created with artificial intelligence.

Recent News

1. Artificial Intelligence Sneaks Into the World Cup Thanks to Google Gemini - WIRED

WIRED | 2026-06-10

Artificial Intelligence Sneaks Into the World Cup Thanks to Google Gemini WIRED

<https://news.google.com/rss/articles/CBMipAFBVV95cUxNTXM5QzJBWEJPZHv0RXQtEVQd3lwazJoOWhY1lpNWxYVEt5N2FvWTNBUR4RWpFbkp1b3Z1bzZBdU5lNjA4Z2pSRk8xRnB4YVEzclZFV1g0cE5kLXRyNXJJUGZ5S3BJWEhDOTVqcmpwd0h2YndzM0t6VIUzOXRVd0RDNTVuOFVvVmpSY2ZzSTNqcEd4U3h5VTFJMFpVl9OUzN4Rg?oc=5>

2. Why Our Obsession With Artificial Intelligence Might Be Completely Wrong - Futura, le média qui explore le monde

Futura, le média qui explore le monde | 2026-06-10

Why Our Obsession With Artificial Intelligence Might Be Completely Wrong Futura, le média qui explore le monde

<https://news.google.com/rss/articles/CBMitAFBVV95cUxPWl9GQlViRFZnekl2Yk5FZjEyaEVCM1FTWVQzX0lwcjVzN00wSmsyQU9HeFixWFdDRWdKdXFuWTdJWkxLSS1vV18yVzBNVUhQUThMYktqeDRoSjdmbXFKdjJaWEp2U2JhNIJUT29WU0w3LVIHdUg3QINpX1dxb2pwUmtFSXJJNjc1ZWtoa2lwZFJaNWhwb0pZQk1tcXc2bC1zX2paTWM5RURtQUtSbEtvMIF3clY?oc=5>

Sources

- 1 [Wikipedia: https://en.wikipedia.org/wiki/Artificial_intelligence](https://en.wikipedia.org/wiki/Artificial_intelligence)
- 2 Artificial Intelligence Sneaks Into the World Cup Thanks to Google Gemini - WIRED: <https://news.google.com/rss/articles/CBMipAFBVV95cUxNTXM5QzJBWEJPZHV0RXQtcEVQd3IwazJoOWhIY1lpNWxYVEt5N2FvWTNBUUR4RWpFbkp1b3Z1bzZBdU5INjA4Z2pSRk8xRnB4YVEzclZFV1g0cE5kLXRyNXJJUGZ5S3BJWEhDOTVqcmpwd0h2YndzM0t6VlUzOXRVd0RDNTVuOFVvVmpSY2ZzSTNqcEd4U3h5VTFJMFpPVI9OUzN4Rg?oc=5>
- 3 Why Our Obsession With Artificial Intelligence Might Be Completely Wrong - Futura, le média qui explore le monde: <https://news.google.com/rss/articles/CBMitAFBVV95cUxPWl9GQlViRFZnekl2Yk5FZjEyaEVCM1FTWVQzX0lwcjVzN00wSmsyQU9HeFixWFdDRWdKdXFuWTdJWkxLSS1vV18yVzBNVUhQUThMYktqeDRoSjdmbXFKdjJaWEp2U2JhNIJUT29WU0w3LVIHdUg3QINpX1dxb2pwUmtFSXJJNjc1ZWtoa2lwZFJaNWWhwb0pZQk1tcXc2bC1zX2paTWM5RURtQUtSbEtvMIF3clY?oc=5>